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Temperature dependence of the refractive index of direct band gap semiconductors near the absorption threshold: Application to GaAs

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Using a semiempirical model of the dielectric constant taking excitonic effects correctly into account, we calculate dn/dT for direct band gap semiconductors by separating the contribution of the excitonic enhancement near the band gap energy E_g from a background term described by the high frequency dielectric constant ϵ_∞ . Comparison of the model with experimental data suggests that $dE_g/dT \approx -4.5 \times 10^{-4}$ eV K⁻¹ and $d\epsilon_\infty/dT \approx 1.32 \times 10^{-4}$ K⁻¹ for GaAs at room temperature, a significant shift from the currently accepted values of these parameters. © 1996 American Institute of Physics. [S0021-8979(96)05018-9]

I. INTRODUCTION

Knowledge of the refractive index n^{1-3} of direct band gap semiconductors in the direct vicinity of the absorption threshold, and of its temperature dependence dn/dT , is crucial in the design and operation of many solid state devices.⁴⁻¹⁴ Not only is the magnitude of dn/dT of interest, but also its energy dependence since devices used in the coupling or modulation of radiation operate at wavelengths corresponding to energies increasingly closer to the band gap E_g , where excitonic effects play a crucial role.¹⁵⁻¹⁸ Therefore, dn/dT is expected to exhibit a strong enhancement in that region. This coefficient has been investigated both experimentally and theoretically for transparent crystals at room temperature. Experiments have clearly shown that in the transparency region and for most crystals, dn/dT is positive and nearly uniform for energies E not too close to the absorption regions,^{2,5} besides, it does not depend on temperature, at least near 300 K. However, the spectral dependence of dn/dT near E_g is still less well known.

The purpose of this work is to improve the knowledge of dn/dT at room temperature, by including a better description of this parameter near the absorption threshold. Because of the form of the experimental results, we will distinguish two terms in dn/dT . The first one emphasizes the excitonic enhancement of dn/dT near E_g , which is mainly controlled by the variation of E_g with temperature, as expected. A semiempirical model for the complex dielectric constant of a direct band gap semiconductor has recently been proposed,¹⁹ in which excitonic effects, and more specifically the Coulomb enhancement factor, are properly taken into account. We use this model in the following calculation. The second term in dn/dT is a background one, corresponding to the almost uniform value of dn/dT in the transparency region, sometimes assimilated to dn_∞/dT (where n_∞ is the square root of the high frequency dielectric constant ϵ_∞). We give in the Appendix a simple way to evaluate dn_∞/dT .

Our expression of dn/dT is then compared with experimental data on GaAs. We are thus led to question the currently accepted values for dE_g/dT and dn_∞/dT at room tem-

perature, which should be increased. Finally, we point out the inadequacy of using linearized expressions of optical parameters, valid between 150 and 300 K, to extrapolate data at temperatures of a few kelvins.

II. THE TWO CONTRIBUTIONS OF dn/dT

The dielectric constant of semiconductors $\epsilon = \epsilon_1 + i\epsilon_2$ is usually split into three terms.^{1,2} The first one corresponds to the reststrahlen contribution due to absorption by the lattice vibrations of photons in the far infrared.^{1,2,12} However, since energies not too far below the absorption threshold are considered, we will discard this contribution. The second term comes from the so-called interband transitions occurring in the vicinity of E_g , while the third term originates with absorption in the fundamental region lying in the near ultraviolet.^{1,2,5,17} Because of the linearity of the Kramers-Krönig relation

$$\epsilon_1(E) - 1 = \frac{2}{\pi} P \int_0^{+\infty} dE' \frac{E' \epsilon_2(E')}{E'^2 - E^2}, \quad (1)$$

we can consider the two contributions $\epsilon_{2,\text{fundamental}}$ and $\epsilon_{2,\text{interband}}$ separately in order to obtain $\epsilon_1(E)$. In the transparency region, $\epsilon_2(E)$ is nearly zero, so that the refractive index defined by

$$n(E) = \text{Re}[(\epsilon_1(E) + i\epsilon_2(E))^{1/2}] \quad (2)$$

is then merely given by $n(E) \approx \sqrt{\epsilon_1(E)}$. Quite generally, the derivative of n with respect to an arbitrary parameter x (which can be temperature, pressure,^{21,22} ...) is given by

$$\frac{dn}{dx} = \frac{1}{\sqrt{\epsilon_1^2 + \epsilon_2^2}} \frac{1}{2n} \left(n^2 \frac{d\epsilon_1}{dx} + \frac{\epsilon_2}{2} \frac{d\epsilon_2}{dx} \right). \quad (3)$$

Consequently, Eq. (3) can be rewritten as

$$\begin{aligned} \frac{dn}{dT} = & \frac{1}{\sqrt{\epsilon_1^2 + \epsilon_2^2}} \frac{1}{2n} \left(n^2 \frac{d\epsilon_{1,\text{fundamental}}}{dT} + n^2 \frac{d\epsilon_{1,\text{interband}}}{dT} \right. \\ & \left. + \frac{\epsilon_{2,\text{interband}}}{2} \frac{d\epsilon_{2,\text{interband}}}{dT} \right). \end{aligned} \quad (4)$$

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In order to highlight the influence of excitonic effects when E is close to E_g , we write dn/dT in a form resembling that proposed by Wherrett *et al.*,¹³ namely,

$$\left. \frac{dn}{dT} \right| = \left. \frac{dn}{dT} \right|_{\text{background}} + \left. \frac{dn}{dT} \right|_{\text{excitonic}}. \quad (5)$$

We first use the complex dielectric constant of Wannier excitons, which reads¹⁹

$$\epsilon_{\text{Wannier}}(E) = \frac{A\sqrt{R}}{(E+i\Gamma)^2} \{g(\xi(E+i\Gamma)) + g(\xi(-E-i\Gamma)) - 2g(\xi(0))\}, \quad (6)$$

$$\xi(z) = \sqrt{\frac{R}{E_g - z}}, \quad (7)$$

$$g(\xi) = 2 \ln \xi - 2\pi \cot(\pi\xi) - 2\psi(\xi) - 1/\xi, \quad (8)$$

where A is the usual coefficient for interband transitions, R the exciton binding energy, Γ the Lorentzian broadening of the transition, and $\psi(z) \equiv (d/dz) \ln \Gamma(z)$ the digamma function. $\epsilon_{\text{Wannier}}$ does not vanish at small energies, but tends to

$$\epsilon_{\text{Wannier}}(0) = \frac{A}{4E_g^{3/2}} + \frac{A\sqrt{R}}{E_g^2}. \quad (9)$$

This small contribution can be included in the background term by defining

$$\epsilon_{\text{exc}}(E) = \epsilon_{\text{Wannier}}(E) - \epsilon_{\text{Wannier}}(0), \quad (10)$$

$$\epsilon_{\text{small } E}(E) = \epsilon_{\text{fundamental}}(E) + \epsilon_{\text{Wannier}}(0), \quad (11)$$

which leads to

$$\left. \frac{dn}{dT} \right| = \left. \frac{dn}{dT} \right|_{\text{small } E} + \frac{1}{\sqrt{\epsilon_1^2 + \epsilon_2^2}} \frac{1}{2n} \left(n^2 \frac{d\epsilon_{1,\text{exc}}}{dT} + \frac{\epsilon_{2,\text{exc}}}{2} \frac{d\epsilon_{2,\text{exc}}}{dT} \right). \quad (12)$$

Actually, the last term on the right hand side of Eq. (12) is negligible with respect to the others in the transparency region or near E_g , so that

$$\left. \frac{dn}{dT} \right| \approx \left. \frac{dn}{dT} \right|_{\text{small } E} + \frac{1}{2n} \frac{d\epsilon_{1,\text{exc}}}{dT}. \quad (13)$$

Equation (13) is the starting point of our discussion about the variation of the refractive index with temperature. Since by construction $\epsilon_{1,\text{exc}}$ vanishes at small energies, $dn/dT|_{\text{small } E}$ prevails at long wavelengths, and corresponds approximately to the measured value of dn/dT in the transparency region—sometimes written as dn_∞/dT —whereas $(2n)^{-1} d\epsilon_{1,\text{exc}}/dT$ is chiefly responsible for the drastic increase of dn/dT observed experimentally just below E_g .^{1,2,5} We must remind here that dn/dT is meaningful only when the temperature excursion ΔT is small enough, and all the more so just below the gap.

In order to validate the present model with available experimental data on GaAs, we will evaluate the two contributions at $E = 1.24$ eV, i.e., at a wavelength of $1 \mu\text{m}$. Talghader and Smith give^{11,23} $dn/dT = (2.67 \pm 0.07) \times 10^{-4} \text{ K}^{-1}$, Mar-

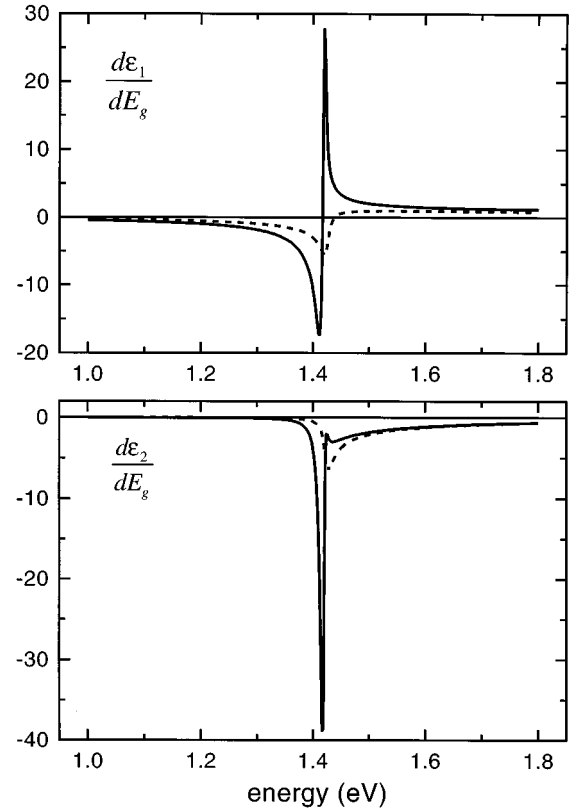


FIG. 1. Derivatives of the real and imaginary parts of ϵ_{exc} with respect to E_g , in the case of GaAs at room temperature, for $A = 2.6 \text{ (eV)}^{3/2}$, $E_g = 1.424 \text{ eV}$, $R = 4.2 \text{ meV}$, and $\Gamma = 6 \text{ meV}$. The dashed curves are obtained when excitonic effects are not included.

ple's data^{24–27} provide an averaged value of about $3 \times 10^{-4} \text{ K}^{-1}$, and Bertolotti *et al.*⁵ find a value slightly higher than $3 \times 10^{-4} \text{ K}^{-1}$.

III. dn/dT NEAR THE ABSORPTION EDGE: THE EXCITONIC ENHANCED CONTRIBUTION

Let us first turn to the contribution of $d\epsilon_{1,\text{exc}}/dT$. It depends on the variation with T of the four parameters A , E_g , R , and Γ given in Eqs. (6)–(8). In the case of GaAs at 300 K, we will use $A = 2.6 \text{ (eV)}^{3/2}$, $E_g = 1.424 \text{ eV}$, $R = 4.2 \text{ meV}$, and $\Gamma = 6 \text{ meV}$. The procedure is straightforward: After the calculation of the derivatives of $\epsilon_{1,\text{exc}}$ with respect to A , E_g , R , and Γ , we must evaluate the derivatives dA/dT , dE_g/dT , dR/dT , and $d\Gamma/dT$. Such an approach has been used in modulation spectroscopy and thermoreflectance,^{28–32} even if mostly applied to the case of the traditional interband transitions in which no excitonic effect is included, and for the E_1 and E_2/E'_0 transitions occurring at higher energies.

It has long been recognized that the prevailing contribution to $d\epsilon_{1,\text{exc}}/dT$ comes from the variation of the energy gap E_g with temperature. Our model confirms this fact and explains the enhancement of dn/dT in the vicinity of the band edge. The spectral dependences of $d\epsilon_{1,\text{exc}}/dE_g$ and $d\epsilon_{2,\text{exc}}/dE_g$ are represented in Fig. 1, along with their corresponding values when no excitonic effect is included.¹³ For the latter curves we have not used the full expression given

in Eq. (25b) of Ref. 13, but have kept its most diverging part only, since the scaling law³³ promoted therein might be of delicate use (n is assumed independent of E_g). Not surprisingly, $d\epsilon_{1,\text{exc}}/dE_g$ and $d\epsilon_{2,\text{exc}}/dE_g$ are quite enhanced by the excitonic contribution. The correlated electron-hole pairs, which are the positive energy excitonic states, are responsible for the Sommerfeld enhancement factor, and thus for the dispersion of n near E_g at room temperature. This dispersion may be traced back to the logarithmic term in Eq. (8): At high energies, the Sommerfeld enhancement factor is different from the well-known $\sqrt{E-E_g}$ expression of the absorption.^{16,19}

We also see that $d\epsilon_{1,\text{exc}}/dE_g$ is negative below the band edge. Since for most semiconductors E_g decreases with temperature, the resulting contribution to dn/dT is positive. One should not forget the influence of the split-off band, whose form is quite similar to that given in Eqs. (6)–(8), with new values for the relevant gap energy $E_g + \Delta$, the exciton binding energy and the increased broadening factor. However, in the case of GaAs, this contribution is but a small fraction (roughly 15% of that related to the direct gap E_g at $E=1.24$ eV, knowing that E_g and $E_g + \Delta$ vary identically with temperature) of the total effect because Δ is quite large ($\Delta \approx 340$ meV). Consequently, we shall not make special mention of this contribution, even though it certainly must be included for materials with small split-off energies Δ . We obtain $d\epsilon_{\text{exc}}/dE_g = (-1.211 - 0.044i) \text{ eV}^{-1}$ at $E=1.24$ eV, which justifies the discarding of the last term in Eq. (12).

Of all the corresponding derivatives of $\epsilon_{1,\text{exc}}$, $d\epsilon_{1,\text{exc}}/dR$ has the largest magnitude: $d\epsilon_{1,\text{exc}}/dR = 17.86 \text{ eV}^{-1}$ at $E=1.24$ eV for GaAs. However, at least for weak binding energies of the exciton, dR/dT is extremely small ($\sim -1.5 \times 10^{-6} \text{ eV K}^{-1}$ in the present case) as can be deduced from the effective masses and their temperature dependence.³⁴ Therefore, this term can be safely omitted in the case of GaAs, and probably for many other semiconductor compounds too.

The variation with temperature of the broadening Γ has been investigated by several groups,^{18,32,35–39} but the ellipsometry and thermorefectance measurements were mainly devoted to the E_1 and E_2/E'_0 transitions which occur at higher energies. Models for $\Gamma(T)$ have been proposed by Gopalan, Lautenschlager, and Cardona,³⁸ as well as Rudin, Reinecke, and Segall.³⁹ Adachi had to introduce the derivative $d\epsilon/d\Gamma$ in order to explain changes in the spectrum of the dielectric constant.³² These models show that $d\Gamma/dT$ is small and positive for the direct gap at room temperature ($d\Gamma/dT \sim 10^{-4} \text{ eV K}^{-1}$). Our model gives $d\epsilon_{1,\text{exc}}/d\Gamma = -0.037 \text{ eV}^{-1}$ at $E=1.24$ eV for GaAs. Consequently, the corresponding contribution to dn/dT is negligible in the spectral range under consideration (this conclusion might not hold in other spectral ranges, for instance near the $E_1/E_1 + \Delta_1$ transitions).

Finally, we consider the variation of A with temperature, which is often neglected even though there exist clear indications that A decreases with T .^{17,40,41} The contribution to dn/dT reads

$$\left. \frac{dn}{dT} \right|_A = \frac{1}{2n} \frac{\partial \epsilon_{1,\text{exc}}}{\partial A} \frac{\partial A}{\partial T} = \frac{1}{2n} \epsilon_{1,\text{exc}} \frac{1}{A} \frac{\partial A}{\partial T}. \quad (14)$$

$A^{-1} dA/dT$ can be estimated by two methods. The first one relates A to the absorption measured just above the gap at different temperatures, using Elliott's formula.^{15,40} The second method is to note that A is proportional to $\mu^{3/2}$, where μ is the reduced mass. Since μ is not too different from the effective mass of the conduction band, we can use Kane's relation^{41,42} to deduce that μ is roughly proportional to E_g , so that

$$\frac{1}{A} \frac{\partial A}{\partial T} \approx \frac{3}{2} \frac{1}{E_g} \frac{\partial E_g}{\partial T}. \quad (15)$$

Both calculations give $dA/dT = -1.2 \times 10^{-3} (\text{eV})^{3/2} \text{ K}^{-1}$ for GaAs at 300 K. Consequently, its contribution to dn/dT amounts to about 20% of the contribution at 1.24 eV originating from $d\epsilon_{1,\text{exc}}/dE_g$ and dE_g/dT , but with the opposite sign since $\epsilon_{1,\text{exc}} > 0$.

The preceding discussion has shown that the prevailing contributions to the excitonic part of dn/dT are the variations of A , E_g and $E_g + \Delta$ with T , and that all these dependences can be related to dE_g/dT . As mentioned above, because two of these contributions nearly cancel each other, only the variation of E_g really matters. If the accepted^{25–27,34,43} values $dE_g/dT = -3.95 \times 10^{-4} \text{ eV K}^{-1}$ and $dn_{\infty}/dT = 1.45 \times 10^{-4} \text{ K}^{-1}$ were used in Eq. (13), we would obtain $dn/dT = 2.1 \times 10^{-4} \text{ K}^{-1}$ at $E=1.24$ eV, which is much less than the expected $3 \times 10^{-4} \text{ K}^{-1}$.

We are thus led to question these two constants, which are clearly too small to account for the experimental data. Let us stress that the above discrepancy would be even larger if excitonic effects were discarded, because of the corresponding decrease of $|d\epsilon_{1,\text{exc}}/dE_g|$, as shown in Fig. 1. In the remaining part of this work, we discuss this problem and propose new values for dE_g/dT and dn_{∞}/dT .

IV. dE_g/dT FOR GaAs AT ROOM TEMPERATURE

The variation of E_g with temperature has been extensively studied in the case of gallium arsenide for which recent photoluminescence or ellipsometry data have been published.^{4,20,44–49} They are respectively, in eV K^{-1} units: -4.35×10^{-4} (Dyment *et al.*⁴⁵), -4.52×10^{-4} (Blakemore²⁰), -5×10^{-4} (Jensen and Torabi⁴), -4.5×10^{-4} (Grilli *et al.*⁴⁶), -4.74×10^{-4} (O'Donnell *et al.*⁴⁷), -5.26×10^{-4} (El Allali *et al.*⁴⁸), -4.37×10^{-4} (Dmitriev⁴⁹).

We adopt the value $dE_g/dT = -4.5 \times 10^{-4} \text{ eV K}^{-1}$. Comparing our model with the experimental data for dn/dT , we see that a very good agreement between experiment and theory can be obtained by using $dn_{\infty}/dT = 2.0$ or $2.1 \times 10^{-4} \text{ K}^{-1}$ instead of $1.45 \times 10^{-4} \text{ K}^{-1}$. Let us now discuss this choice, and see how this new value compares with other experimental data.

V. dn/dT IN THE TRANSPARENCY REGION: A DISCUSSION

The high frequency dielectric constant ϵ_{∞} describes the approximately uniform value of $\epsilon_1(E)$ for energies well be-

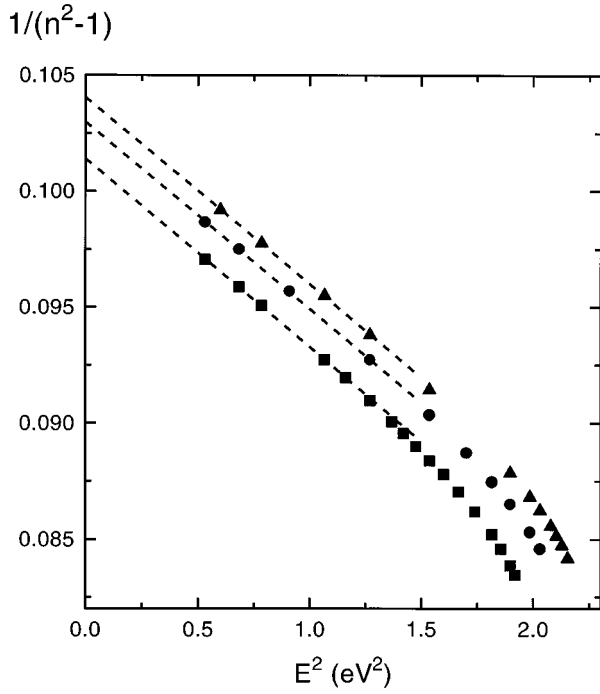


FIG. 2. Determination of n_{∞} from Marple's data measured at three temperatures ($\blacktriangle$ 103 K $\bullet$ 187 K $\blacksquare$ 300 K), the asymptotic value at $E=0$ of the dashed lines being $1/(n_{\infty}^2-1)$.

low E_g but still above the reststrahlen region. It can be viewed as the limit when E goes to zero of $\epsilon_1(E)$ in Eq. (1), in which the contribution to ϵ_2 caused by the absorption of light by lattice vibrations has been removed, so that

$$\epsilon_{\infty} = n_{\infty}^2 = 1 + \frac{2}{\pi} P \int_0^{+\infty} dE' \frac{\epsilon_2(E')}{E'}. \quad (16)$$

By expanding Eq. (1) at low energies one finds

$$\epsilon_1(E) \approx \epsilon_{\infty} + E^2 \frac{2}{\pi} P \int_0^{+\infty} dE' \frac{\epsilon_2(E')}{E'^3} + \dots, \quad (17)$$

which is often expressed as a simple Sellmeier formula, or alternatively as the [1,0] Padé approximant^{18,50} of Eq. (1):

$$\epsilon_1(E) - 1 \approx \frac{a}{b - E^2}. \quad (18)$$

Equation (18) lends itself very nicely to a linear regression,³ namely,

$$\frac{1}{\epsilon_1(E) - 1} = \frac{b}{a} - \frac{E^2}{a}, \quad (19)$$

as can be seen on Fig. 2, which represents Marple's data at three different temperatures. A plot of $[n^2(E) - 1]^{-1}$ vs E^2 provides therefore a , b , and $\epsilon_{\infty} = 1 + a/b$. Equation (18) is valid at small energies only, and fails when E nears the band edge. One can note from Fig. 2 that the slopes at origin of the three different curves are nearly the same, so that a is almost independent of temperature. Experimentally, n can be determined below the gap by the prism method^{5,24} or by studying the eigenmodes of optical wave guides.^{10,51}

Let us mention here the theoretical work devoted to the determination of dn/dT in the transparency region. Moss proposed a relationship⁵² between n_{∞} and E_g

$$n_{\infty}^4 E_g = \text{constant}, \quad (20)$$

which gives reasonable agreement with experiments for a great variety of crystals. However, Cardona⁴² showed it cannot be used to calculate dn_{∞}/dT . Theoretical models on the variation of n_{∞} with temperature adopt a semiempirical approach,^{2,4-7} in which a few parameters vary with T . More elaborate calculations predict $dn/dT = 1.84 \times 10^{-4} \text{ K}^{-1}$ (Ref. 8) or $1.9 \times 10^{-4} \text{ K}^{-1}$ (Ref. 9). We give in the Appendix a quick way to obtain a reasonable estimate of this quantity.

If we now turn back to experiments, values of n_{∞} can be deduced from Marple's data²⁴ taken at three different temperatures, using Afromowitz's fit as in Fig. 2. They give an averaged $\Delta n_{\infty}/\Delta T$ equal to $1.87 \times 10^{-4} \text{ K}^{-1}$ between 103 and 187 K, and $2.05 \times 10^{-4} \text{ K}^{-1}$ between 187 and 300 K, which corresponds to a much smoother variation (with respect to T) of dn_{∞}/dT than the variation deduced by Seraphin and Bennett.²⁵ This is not unexpected since all energy gaps vary almost linearly with T above 150 K. Moreover, if we consider n between 103 and 300 K at a wavelength of $1.4 \mu\text{m}$ where the band edge influence is still small (roughly $2 \times 10^{-5} \text{ K}^{-1}$), we find $\Delta n/\Delta T = 2.18 \times 10^{-4} \text{ K}^{-1}$. The value $dn_{\infty}/dT = 1.9 \times 10^{-4} \text{ K}^{-1}$ is also given by Pikhtin and Yas'kov² between 15 and 35 °C. Therefore, the value we propose at the end of Sec. IV agrees with experimental data.

Further useful information could be obtained from measurements performed at low temperatures (down to 40 K), as described by Kisting *et al.*⁵¹ or mentioned by Stern.¹ dE_g/dT is much lower at these temperatures because the variation with T of energy gaps obeys Varshni's law^{20,43} or equivalent expressions.⁴⁷ Indeed, dE_g/dT is roughly equal to $-1.7 \times 10^{-4} \text{ eV K}^{-1}$ at 40 K which is 2.6 times smaller than the 300 K value. dn_{∞}/dT is reduced accordingly, so that one has to get closer to the absorption threshold to observe significant values of dn/dT . In the 40–80 K temperature range and at $E = 1.4 \text{ eV}$, i.e., not too far from the band edge because $E_g = 1.513 \text{ eV}$ at 40 K, Kisting *et al.*⁵¹ found $dn/dT = 3.2 \times 10^{-4} \text{ K}^{-1}$. Their result is, again, consistent with our calculations.

Having estimated $dn_{\infty}/dT \approx 2 \times 10^{-4} \text{ K}^{-1}$ at room temperature and stressed on the variations of n_{∞} and E_g at low temperatures, we would like to address Blakemore's proposal²⁰ for $\epsilon_{\infty}(T)$ and stress his words of caution. As explained above, we expect this variation to vanish at very low temperatures.¹⁴ Consequently, a linearized expression of $\epsilon_{\infty}(T)$, which can be justified in the range 100–300 K, cannot be used down to 2 K.⁵³ From its room temperature value and $dn_{\infty}/dT \approx 2 \times 10^{-4} \text{ K}^{-1}$, we deduce $d\epsilon_{\infty}/dT \approx 1.32 \times 10^{-4} \text{ K}^{-1}$ (see also the value $1.24 \times 10^{-4} \text{ K}^{-1}$ given in Ref. 14) or, equivalently,

$$\epsilon_{\infty}(T) \approx 10.48(1 + 1.26 \times 10^{-4} T). \quad (21)$$

Obviously, Eq. (21) gives a variation of the high frequency dielectric constant with temperature much greater than Blakemore's. Nevertheless, this increased value is not inconsistent with some other experimental data, quite on the

contrary. The static and high frequency dielectric constants are linked by the well-known Lyddane–Sachs–Teller relation

$$\epsilon_0(T) = \epsilon_\infty(T) \left[\frac{\omega_{\text{LO}}(T)}{\omega_{\text{TO}}(T)} \right]^2, \quad (22)$$

where $\omega_{\text{LO}}(T)$ and $\omega_{\text{TO}}(T)$ are the long wavelength optical longitudinal and transverse mode phonon frequencies. Accurate measurements of these frequencies as functions of T have been performed using Raman measurements.⁵⁴ They show without ambiguity that these two quantities decrease at the same rate of $1.35 \times 10^{-2} \text{ cm}^{-1} \text{ K}^{-1}$. From this determination, we deduce that between 150 and 450 K

$$\left[\frac{\omega_{\text{LO}}(T)}{\omega_{\text{TO}}(T)} \right]^2 \propto (1 + 0.78 \times 10^{-5} T), \quad (23)$$

where the temperature coefficient is significantly smaller than in Ref. 20, the small temperature coefficient originating from the different phonon energies. From $\epsilon_0 = 12.9$ at 300 K²⁰ and Eqs. (21)–(23), we obtain

$$\epsilon_0(T) \approx 12.4(1 + 1.35 \times 10^{-4} T), \quad (24)$$

which is valid only near room temperature, of course. The asymptotic value at $T = 0 \text{ K}$, namely, 12.4, turns out to be the same as Blakemore's. However, this value should be considered as a *lower bound* to the actual value $\epsilon_0(T = 0 \text{ K})$.¹⁴ We see that the temperature coefficients deduced from the data of Champlin and Glover⁵⁵ between 100 and 300 K and from Eq. (24) are nearly identical. We consider this similarity as an additional, even if indirect, indication that $d\epsilon_\infty/dT$ is larger than its previously accepted value. One may note that this proposed increase also reduces the discrepancy between the theoretical models of Refs. 8 and 9, and experimental results. Finally, let us recall that the expressions given above apply for semi-insulating GaAs. For doped semiconductors, n_∞ and dn_∞/dT would be reduced because of free carrier absorption.

VI. CONCLUSION

We have presented a model for the derivative of the refractive index with respect to the temperature that provides a more accurate description of this coefficient near the band edge for a direct band gap semiconductor. This model has been successfully compared to experimental data for gallium arsenide. This comparison has shown that the room-temperature values of dE_g/dT and dn_∞/dT found in tables should be modified by a slight amount, in order to be in agreement with other published data. New low-temperature (below 50 K) measurements of n in the transparency region would be very interesting because they could lead to expressions of $\epsilon_\infty(T)$ [and therefore of $\epsilon_0(T)$] valid beyond the usual range 150–300 K.

Finally, the present model could be helpful in the design of devices based on direct gap semiconductors, for which experimental data is scarce or even not yet available. It could also be easily adapted for the real-time thickness control by reflectometry of the growth of semiconductor samples, at the corresponding high temperatures.

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APPENDIX

We propose here an expression for dn_∞/dT , using a very simple phenomenological approach. Because the imaginary part of the dielectric constant is essentially coming from the fundamental absorption region, and because the main effect of the temperature change is a quasi rigid band shift $-\alpha$ (with $\alpha > 0$ for most semiconductor compounds), we assume that

$$\epsilon_2(E, T + dT) \approx \epsilon_2(E + \alpha dT, T). \quad (A1)$$

This neglects the thermal expansion of the crystal, whose influence amounts to a fraction of the contribution treated here. Inserting Eq. (A1) in the Kramers–Krönig relations, one easily finds

$$\frac{\partial \epsilon_\infty}{\partial T} = \alpha \frac{2}{\pi} P \int_0^{+\infty} dE' \frac{\epsilon_2(E')}{E'^2}. \quad (A2)$$

We can write

$$\begin{aligned} \frac{2}{\pi} P \int_0^{+\infty} dE' \frac{\epsilon_2(E')}{E'^2} \\ = \frac{1}{E_0} \frac{2}{\pi} P \int_0^{+\infty} dE' \frac{\epsilon_2(E')E'}{E'^2}, \end{aligned} \quad (A3)$$

where E_0 corresponds to an averaging of $\epsilon_2(E')$ and can be evaluated in the case of diamond- or zincblende-structured semiconductors by the simple formula

$$E_0 \approx (E_X + E_\Lambda)/2, \quad (A4)$$

E_X and E_Λ being the energies of the peaks in the ϵ_2 spectrum. dn_∞/dT is then given by

$$\frac{dn_\infty}{dT} \approx \frac{n_\infty^2 - 1}{2n_\infty} \frac{\alpha}{(1/2)(E_X + E_\Lambda)}. \quad (A5)$$

An approximation for dn_∞/dT can be obtained easily from the knowledge of α , which is mainly given by the energy shifts of E_X and E_Λ with temperature (and not that of E_g !). The agreement with Bertolotti *et al.*'s data⁵ on GaAs, InAs, etc. is satisfactory, given the oversimplified assumptions leading to Eq. (A5).

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